2 **(a)** The first law of thermodynamics states that the increase in *internal energy* of a *closed* system is the sum of the *heat* supplied and work done on the system.

Explain what is meant by

(i) *internal energy*,

……

……

……

…….. [1]

(ii) *closed* system,

……

…….. [1]

(iii) *heat*.

……

…….. [1]

(b) **(i)** An ideal gas occupies a volume V at a pressure p and a temperature of T inside an insulated container as shown in Fig. 2.1. It is held in place by a frictionless piston. The other section of the container contains a vacuum.

Using the first law of thermodynamics, explain why the temperature of the gas remains constant when the piston is no longer held in place and the gas is allowed to expand to occupy the entire volume of the container.

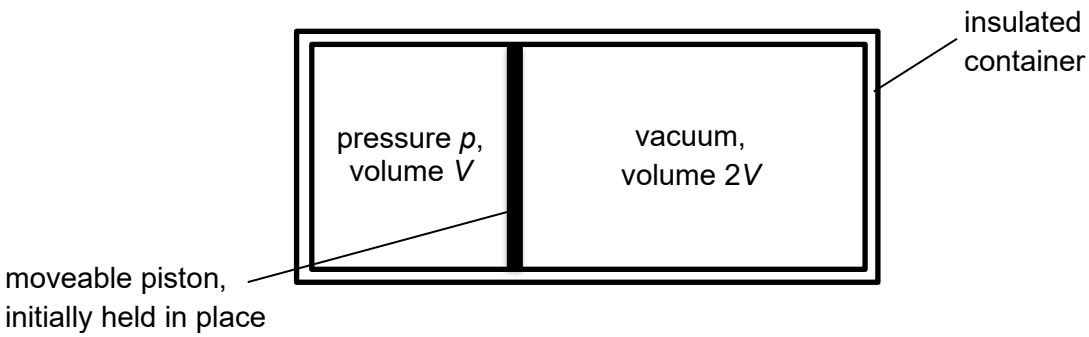

Fig. 2.1

……

……

……

…….. [3]

(ii) Hence or otherwise, determine the ratio of the initial pressure to the final pressure of the gas.

ratio = ... [2]

(c) In 1816, Robert Stirling patented the Stirling engine which operates by cyclic compression and expansion of air or other gas at different temperatures, such that there is a net conversion of thermal energy to mechanical work.

Figure 2.2 shows the cycle of changes that a gas undergoes in the Stirling engine.

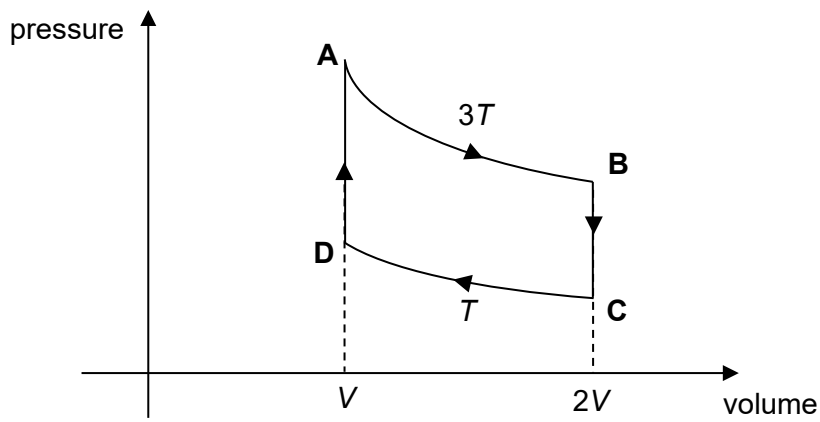

Fig. 2.2

Fig. 2.3 shows the processes that are involved in the cycle of changes.

section of cycle	description	heat supplied
A → B	increase of volume at constant temperature	
B → C	decrease of pressure at constant volume	
C → D	decrease of volume at constant temperature	
D → A	increase of pressure at constant volume	

Fig. 2.3

(i) Complete the table in Fig. 2.3 by writing 'to' or 'from'. [2]

(ii) Determine, in terms of n, R and T, the change in internal energy for n mol of a monatomic ideal gas for the change D → A.

[1]

(iii) The work done on the gas for a change in volume at constant temperature can be given by the expression $W = n\,R\,T\,ln\frac{V_{initial}}{V_{final}}$.

Hence or otherwise, derive an expression for the net heat transferred to the gas during one cycle in terms of n, R and T. Explain your working clearly.

[3]

(iv) The efficiency of the Stirling engine shown is determined by the net work done by the system divided by the heat transferred to the system. Using your working in **(c)(i)** to **(iii)**, determine the efficiency.

efficiency = …… [2]